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STRUCTURAL ELUCIDATION OF A NEUTRAL WATER-SOLUBLE α-D-GLUCAN FROM THE FUNGUS OF *HERICIUM ERINACEUS*

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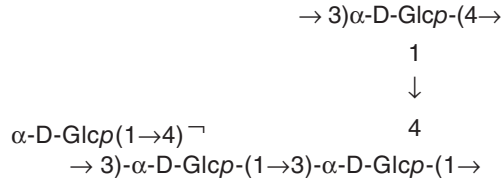
ABSTRACT

A new α-D-glucan (HEPF5), with a molecular weight of 4.23×10^5 Da as determined by high-performance liquid chromatography, was isolated from the water extract of the fruiting bodies of *Hericium erinaceus*. Its chemical structure was characterized by sugar and methylation analysis along with infrared (IR), ^{1}H and ^{13}C nuclear magnetic resonance spectroscopy, including correlated spectroscopy, total correlation spectroscopy, heteronuclear multiple quantum correlation spectroscopy and nuclear Overhauser enhancement spectroscopy experiments for linkage and sequence analysis. The polysaccharide is composed of a multibranch tetrasaccharide repeating unit with the following structure:



PRACTICAL APPLICATIONS

Mushroom-derived polysaccharides have emerged as an important class of bioactive substances, to which numerous medicinal and therapeutic properties have been attributed. *Hericium erinaceus* is a traditional Chinese medicinal fungus distributed throughout China. It is used to treat gastric ulcers, chronic gastritis and other digestive tract-related diseases. As the function is intimately related to the structure, characterization of highly complex and diverse polysaccharides is highly desirable. Despite the large amount of knowledge on the structure of polysaccharide, elucidation of the fine structures of heteropolysaccharide from varied sources and diverse conditions still remains as a matter of immense interest and importance. This paper describes the structure of a novel water-soluble heteropolysaccharide isolated from the fruiting bodies of *H. erinaceus*. It should help in correlating the structure and function relationship, which is significant for its utilization both in pharmaceutical and food industries.

INTRODUCTION

The basidiomycete, *Hericium erinaceus*, is a traditional Chinese medicinal fungus distributed throughout China. It is used to treat gastric ulcers, chronic gastritis and other digestive tract-related diseases. Both the fruiting bodies and the fungal mycelia have been reported to contain bioactive polysaccharides (Zhou *et al.* 1991) that are reputed to exhibit various pharmacological activities including enhancement of the immune system, as well as antitumor, hypoglycemic and anti-aging properties (Yang *et al.* 2000; Nie and Zhu 2003). Glucan has been extracted from many basidiomycetous fungi and showed many bioactivities, such as immunomodulatory activity (Molinaro *et al.* 2000; Dong *et al.* 2006) and antitumor effect (Jong *et al.* 2009) by stimulating natural killer cells, T cells, B cells and macrophage-dependent immune system

responses. In order to identify the correlations between structural and studies on polysaccharides from *H. erinaceus*, in this paper, the structural elucidation of HEPF5, a novel neutral polysaccharide purified from the fruiting bodies of *H. erinaceus*, is described.

MATERIALS AND METHODS

Materials and Chemicals

Fruiting bodies were purchased from Pan'an in Zhejiang Province, China, and identified by Professor Taihui Li in Guangzhou Institute of Microbiology. DEAE-Sepharose Fast Flow and Sephacryl S500 High Resolution were purchased from Amersham Pharmacia Biotech (Little Chalfont, Bucks, U.K.). Dextrans and the monosaccharides, D-Gla, D-Ara, L-Fuc, L-Rha, D-man, D-Xyl and D-Glc, were from Sigma. All other reagents were of A.R. grade and made in China. Gas chromatography–mass spectrometry (GC-MS) was carried out using a ThermoFinnigan TRACE MS (ThermoFinnigan Co., Silicon Valley, CA), and nuclear magnetic resonance (NMR) spectra were determined with a Varian INOVA 500 (International Equipment Trading Ltd., Vernon Hills, IL).

Isolation and Purification of Polysaccharide

The total fruiting bodies of *H. erinaceus* were first exhaustively extracted with 95% ethanol under reflux for 12 h to remove lipids and other minor molecular impurity. This step was repeated three times. After filtration, the residue was dried by air at room temperature, and then extracted with boiling distilled water thrice (2 h for each); the liquid extracts were collected together. After the total aqueous filtrate was concentrated into one-tenth of the original volume, 95% ethanol was added into the aqueous filtrate until the final alcohol concentration reached 30%; the precipitate was separated, defined as HEPF30. A portion of HEPF30 was dissolved in water, the insoluble residue removed by centrifugation. The supernatant was applied to a DEAE-Sepharose Fast Flow column (XK26 × 100 cm), eluted first with water and then with 0–2 M gradient NaCl. The elute was collected by an autocollector and detected by means of the phenol–sulfuric acid method (Zhang 1999). HEPF5 was purified by gel permeation chromatography on a column of Sephacryl S-500 High Resolution (XK26 × 100 cm) from the water elute of HEPF30.

Determination of Purity and Molecular Weight

Determination of homogeneity and molecular weight of samples was done by high-performance liquid chromatography (HPLC) on a linked column of TSK PWXL 4000 and 3000 gel filtration columns, eluting with 0.1 M phosphate buffer solution and 0.3 M $NaNO_3$ at pH 7.0 with a flow rate of 0.6 mL/min. The column was kept at 30.0 ± 0.1C. The column was calibrated by dextrans (T-700, 580, 300, 110, 80, 70, 40, 9.3, 4) using linear regression. All samples were prepared as 0.2 % (w/v) solutions, and 10 μL of solution was analyzed in each run.

Spectroscopic Methods

The Fourier transform infrared spectrum of the polysaccharide was measured on a Perkin-Elmer (Wellesley, MA) 599B Fourier Transform Infrared spectroscopy spectrometer using the KBr disk method.

Monosaccharide Composition Analysis

HEPF5 (2 mg) was hydrolyzed with 2 M trifluoroacetic acid (TFA) at 110C for 2 h, and the monosaccharide composition was determined by high-performance anion-exchange chromatography (HPAEC) using a Dionex LC30 (Dionex Corporation, Sunnyvale, CA) equipped with a CarboPac PA20 column (3 mm × 150 mm). The column was eluted with 2 mM NaOH (0.45 mL/min) and the monosaccharides were monitored using a pulsed amperometric detector (Dionex) (Yang *et al.* 2005).

Methylation Analysis

Vacuum-dried polysaccharide (2 mg) was dissolved in dimethyl sulfoxide (DMSO) (2 mL) and methylated by treatment with NaOH-DMSO (0.2 mL) suspension and methyliodide (0.2 mL) by the method of Kalyan and Paul (1992). The reaction mixture was extracted with trichloromethane, and the trichloromethane was then removed by evaporation. Complete methylation was confirmed by the disappearance of the OH band (3,200 ~ 3,700/cm) in the infrared spectrum. The permethylated polysaccharide was hydrolyzed by treatment with formic acid (88%, 0.5 mL), H_2O (0.1 mL) and TFA (0.05 mL) for 16 h at 100C. The partially methylated sugars in the hydrolysate were reacted with sodium borohydride and acetylated by acetic anhydride, and the resulting mixture of alditol acetates was analyzed by GC-MS using a DB-5 column (30 m × 0.25 mm × 0.25 μm) and a temperature program consisting of 80C to 200C at 5C/min, increasing to 215C at 2C/min, and finally to 270C at 20C/min.

NMR Analysis

HEPF5 (30 mg) was lyophilized three times in D_2O (0.5 mL). The ^1H NMR and ^{13}C NMR (60C) spectra were determined in 5-mm tubes using a Varian INOVA 500 NMR spectrometer. ^1H chemical shifts were referenced to the water resonance at δ 4.41 at 60C as internal standard. ^{13}C chemical shifts and were deter-

mined in relation to 4,4-dimethyl-4-silapentane-1-sulfonic acid (δ 0.00 ppm) calibrated externally. DEPT-135 spectroscopy, ^{1}H–^{1}H correlated spectroscopy (COSY), total correlation spectroscopy (TOCSY) and heteronuclear multiple quantum correlation spectroscopy (HMQC) were used to assign signals. Two-dimensional heteronuclear multiple-bond correlation spectroscopy (HMBC) and two-dimensional nuclear Overhauser enhancement spectroscopy (NOESY) were used to assign interresidue linkages and sequences.

RESULTS

A novel polysaccharide was extracted and purified from the fruiting bodies of *H. erinaceus* by hot-water extraction, ethanol precipitation and anion-exchange and gel-filtration chromatography subsequently. The water-soluble polysaccharide termed HEPF5 was purified using high-resolution Sephacryl S-500 gel-permeation chromatography. The HPLC profile showed a single and symmetrical peak, indicating it was a homogeneous polysaccharide (Fig. 1) and its average molecular weight of 4.23×10^5 Da.

HEPF5 is a white powder that is soluble in water. Lack of absorption at 280 and 260 nm by ultraviolet scanning showed that the polysaccharide contained no phenolic compounds, protein or nucleic acid. Sugar compositional analysis of HEPF5 determined by HPAEC indicated that it was composed of only D-glucose (Fig. 2).

The infrared (IR) spectrum of HEPF5 is shown in Fig. 3. The broadly stretched intense peak at 3,423.1/cm was due to the hydroxyl stretching vibration of the polysaccharide. The band at 2,927.5/cm was due to C–H stretching vibration. The relatively strong absorption peak at 1,646.2 and 1,413.6/cm also indicated the characteristic IR absorption of polysaccharides. Two stretching peaks at 1,153.2 and 1,024.0/cm in the IR spectrum suggested the presence of C–O bonds (Zhao *et al.* 2007). No absorption peaks at 1,730/cm indicated that there were no uronic acids.

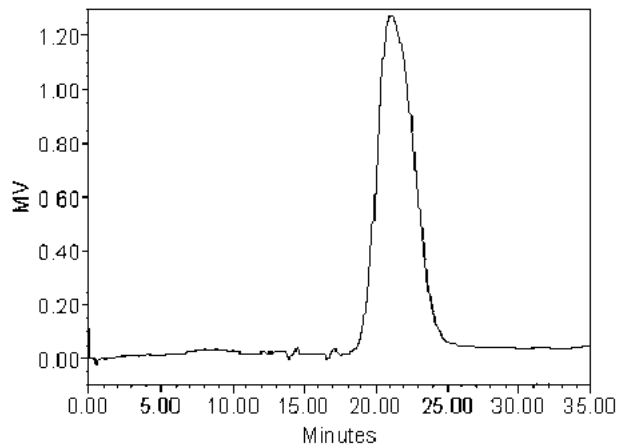


FIG. 1. HIGH-PERFORMANCE LIQUID CHROMATOGRAPH ELUTION OF HEPF5 POLYSACCHARIDE ISOLATED FROM THE FRUITING BODIES OF *H. ERINACEUS*

The interglycosidic linkages between monosaccharide residues of HEPF5 were investigated by methylation analysis. The polysaccharide was methylated twice and after acid hydrolysis, methylated sugars were converted to partially methylated alditol acetates. As a result of methylation analysis, 2,3,4,6-tetra-O-methylglucose, 2,6-di-O-methylglucose were detected (Table 1).

The ^{1}H NMR spectrum of the polysaccharide contained signals for two anomeric protons at δ 5.03–5.44, other sugar protons in the region of δ 3.50–4.26 (Fig. 4). The ^{13}C DEPT-135 NMR spectrum of the polysaccharide contained signals for one major anomeric carbon at δ 102.7 (Fig. 5) and a small

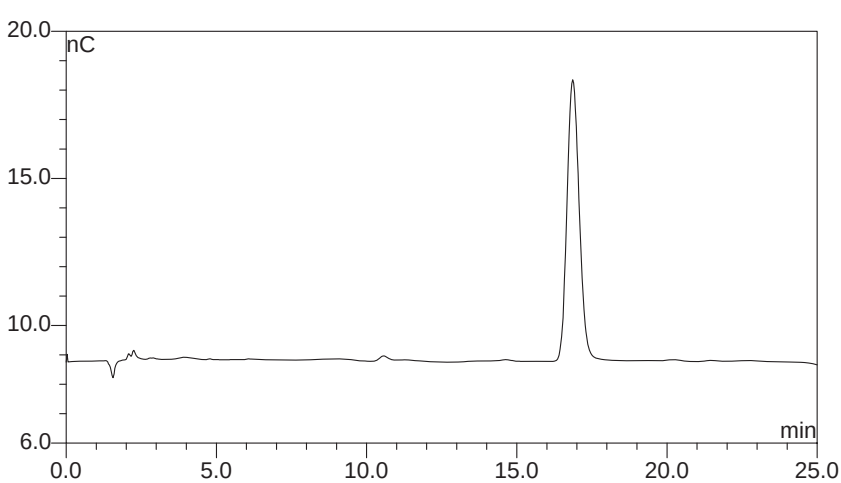


FIG. 2. HIGH-PERFORMANCE ANION-EXCHANGE CHROMATOGRAPHY OF THE MONOSACCHARIDE IN HEPF5 POLYSACCHARIDE ISOLATED FROM THE FRUITING BODIES OF *H. ERINACEUS*

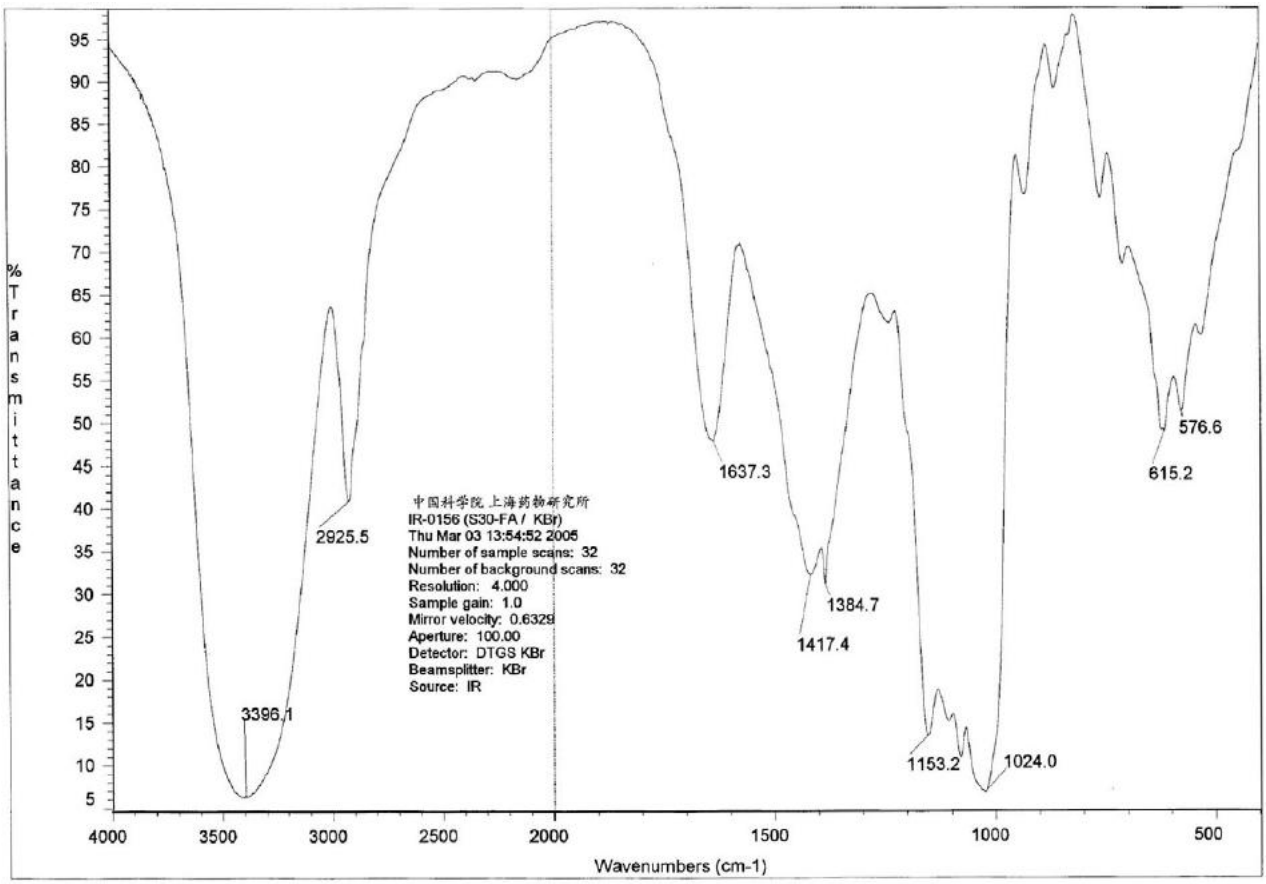


FIG. 3. INFRARED (IR) SPECTRUM OF HEPF5 POLYSACCHARIDE ISOLATED FROM THE FRUITING BODIES OF *H. ERINACEUS*

signal for anomeric carbons at δ 100.8, validated by the HMQC spectrum. CH_2 groups (C-6 of Glc) with one reverse peaks at δ 63.5 in DEPT-135 spectrum and other sugar ring carbons linked to oxygen in the region of δ 67.2–80.3.

The ^{1}H NMR and ^{13}C NMR spectra of the polysaccharide were assigned using ^{1}H–^{1}H COSY, TOCSY, ^{1}H–^{13}C HMQC and HMBC experiments (Table 2). All the ^{1}H resonance for residue a and b were readily assigned from the ^{1}H–^{1}H COSY spectrum and confirmed from the TOCSY spectrum. Magnetization relayed well through the spin system, as expected for the gluco-configuration, and all cross peaks were clearly visible. ^{13}C resonances were assigned from the HMQC spectrum. H-1 of residue a and b appears as a singlet ($J_{H\text{-}1,H\text{-}2}$ <3 Hz) and H-1 of in the ^{1}H NMR spectrum and its chemical shifts are more than 5.00 ppm showed that residue a and b has a α-configuration and validated by H-1 and H-2 intraresidue correlations presented in the NOESY spectrum. Thus, a and b were identified as α-D-Glc*p*.

Comparison of the downfield chemical shifts data for C-3 (δ 80.3) and C-4 (δ 76.1) of the residue a with respect to those reported for glycosides permitted (Agrawal 1992) indicated that residue a is 3, 4-linked α-D-Glc*p* and residue b is 1-linked α-D-Glc*p*.

The sequence of the residues in the repeating unit was established from NOESY spectrum, which showed clear interresidue NOEs between H-1 of residue b and H-4 of residue a, H-1 of residue a and H-3, H-4, and NOE correlations data in Table 3.

TABLE 1. GC-MS DATA FOR METHYLATION ANALYSIS OF HEPF5

Methylated sugars	Linkages types	Molar ratios (mol %)	Major mass fragments (*m/z*)
2,3,4,6-Me_4-Glc*p*	1,-linked Glc*p*	1.0	43,71,87,101,117,129,145,161,205
2,3,6-Me_3-Glc*p*	1,3,4-linked Glc*p*	2.98	43,71,87,101,117,129,143,231,305

GC-MS, Gas chromatography–mass spectrometry.

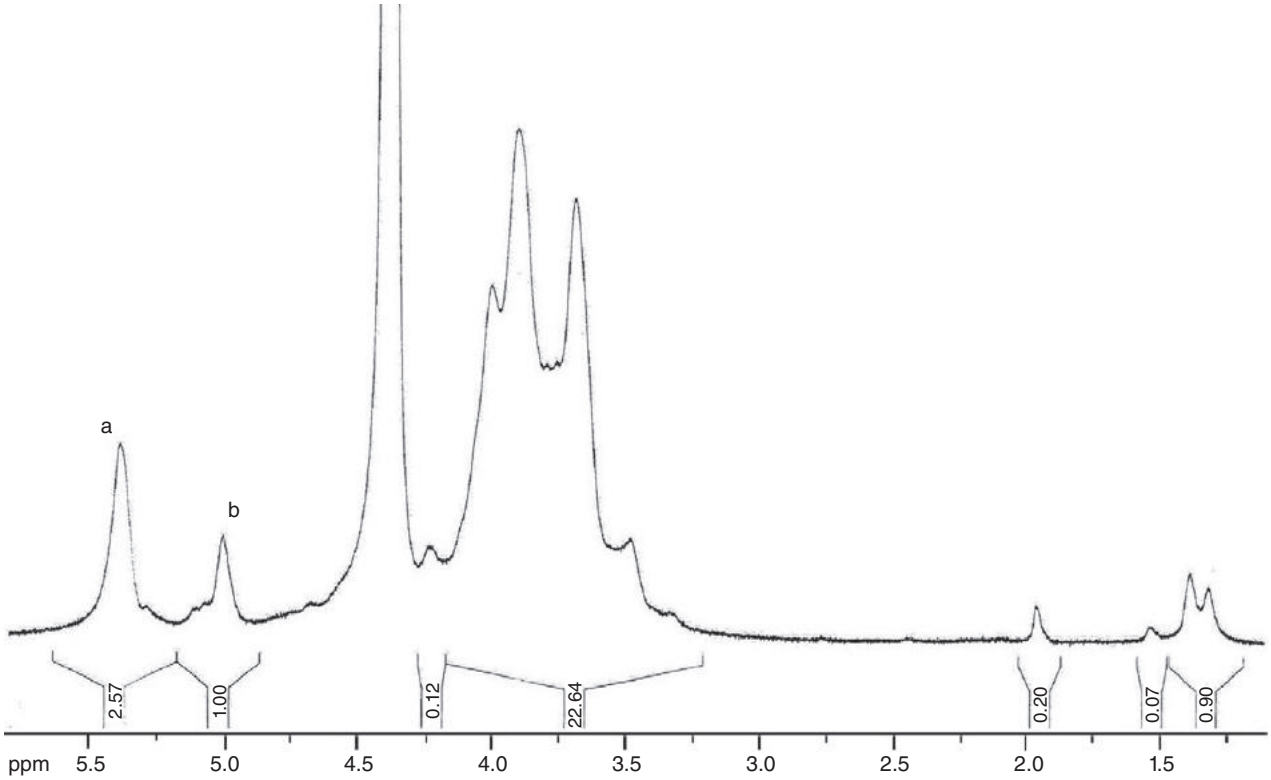


FIG. 4. 500 MHZ ^{1}H NUCLEAR MAGNETIC RESONANCE SPECTRUM OF HEPF5 POLYSACCHARIDE AT 60C ISOLATED FROM THE FRUITING BODIES OF *H. ERINACEUS*

The combined chemical and NMR data permit the structure of the mainly repeating units for HEPF5 with the following structure:

→ 3)α-D-Glc*p*-(4→
1
↓
α-D-Glc*p*(1→4) ¬ 4
→ 3)-α-D-Glc*p*-(1→3)-α-D-Glc*p*-(1→

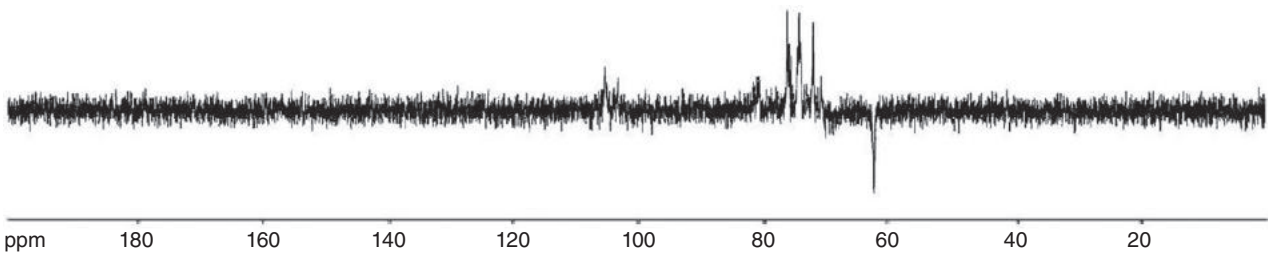


FIG. 5. ^{13}C-DEPT-135 NUCLEAR MAGNETIC RESONANCE SPECTRUM OF HEPF5 POLYSACCHARIDE ISOLATED FROM THE FRUITING BODIES OF *H. ERINACEUS*

TABLE 2. CHEMICAL SHIFTS DATA FOR HEPF5 POLYSACCHARIDE ISOLATED FROM THE FRUITING BODIES OF *H. ERINACEUS*

		Proton or carbon						
Residue		1	2	3	4	5	6a	6b
→3,4)- α-D-Glc*p* (a)	H	5.44	3.69	3.72	4.03	3.90	3.95	3.95
	C	102.7	74.5	80.3	76.1	74.8	63.5	
α-D-Glc*p* (b)	H	5.03	3.92	4.11	3.69	3.80	3.95	3.95
	C	100.8	74.4	71.2	74.5	76.1	63.5	

TABLE 3. INTERGLYCOSIDIC CORRELATIONS FROM NOESY SPECTRA OF HEPF5 FROM THE FRUITING BODIES *H. ERINACEUS*

Residue	Proton	NOE to *
→3,4)-α-D-Glc*p* (**a**)	H-1	3.69 (**a**; H-2), 3.72 (**a**; H-3), 4.03 (**a**; H-4)
α-D-Glc*p* (**b**)	H-1	5.03 (**b**; H-2), 4.03 (**a**; H-4)

* Interresidue NOEs are shown in bold fond.
NOESY, nuclear Overhauser enhancement spectroscopy.

DISCUSSION

In most of the fungi examined, polysaccharides composed of β-glucans, α-D-manmans, have been reported to be the major components of the cell wall and the intercellular matrix, with the latter found mainly in yeast cell walls and medicinal mushrooms (Vingradov *et al.* 1998). In contrast, polysaccharide consisting of fucogalactan (Zhang *et al.* 2006), fucoglucogalactan (Zhang *et al.* 2007) and β-glucan (Dong *et al.* 2006) have been found in the fungus *H. erinaceus*. The biological effects of these polysaccharides have been widely studied for their immunostimulating and antitumor activities (Tokunka *et al.* 2000). However, structure containing a multibranch α-D-Glucan has not been previously reported in the *H. erinaceus*, and HEPF5 is therefore a novel fungal polysaccharide.

ACKNOWLEDGMENTS

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REFERENCES

AGRAWAL, P.K. 1992. NMR spectroscopy in the structure elucidation of oligosaccharides and glycosides. Phytochemistry. *31*, 3307–3350.

DONG, Q., JIA, L.M. and FANG, J.N. 2006. A β-D-glucan isolated from the fruiting bodies of *Hericium erinaceus* and its aqueous conformation. Carbohydr. Res. *341*, 791–795.

JONG, S.E., JAE, Y.C. and EOCK, K.H. 2009. Study on macrophage activation and structural characteristics of purified polysaccharides from the liquid culture broth of *Hericium erinaceus*. Carbohydr. Res. *78*, 162–168.

KALYAN, R.A. and PAUL, B.T. 1992. A comprehensive procedure for preparation of partially methylated alditol acetates from glycoprotein carbohydrates. Anal. Biochem. *203*, 101–108.

MOLINARO, A., LANZETTA, R., MANCINO, A., EVIDENTE, A., ROSA, M.D. and INANRO, A. 2000. Immunostimulant (1→3)-D-glucans from the cell wall of *Cryphonectria parasitica* (Murr.) Barr strain 263. Carbohydr. Res. *329*, 441–445.

NIE, J.S. and ZHU, S.F. 2003. *Hericium erinaceus* polysaccharide and its antineoplastic activity and influence on immunity on mice. J. Shanxi Med. *32*, 107–109.

TOKUNKA, K., OHNO, N., ADACHI, Y., TANAKA, S., TAMURA, H. and YADOMAE, T. 2000. Immunopharmacological and immunotoxicological activities of a water-soluble (1→3)-beta-D-glucan, CSBG from *Candida* spp. Int. J. Immunopharmacol. *22*, 382–394.

VINGRADOV, E., PETERSEN, B. and BOCK, K. 1998. Structural analysis of the intact polysaccharide manman from *Saccharomyces cerevisiae* yeast using ^{1}H and ^{13}C NMR spectroscopy at 750 MHz. Carbohydr. Res. *307*, 177–185.

YANG, Y., YAN, H.F., WANG, C.G. and BAI, Y.Q. 2000. Studies on immunoregulatory function of *Hericium erinaceus*. Acta Edulis Fungi *7*(1), 19–22.

YANG, R.Z., ZHANG, J.S. and TANG, Q.J. 2005. High performance anion exchange chromatography method to determine the monosaccharide composition of polysaccharide. Edilbe Fungi China *24*, 42–44.

ZHANG, W.J. 1999. *Biochemical Techniques in Complex Carbohydrates*, p. 11, Zhejiang University Press, Hangzhou, China.

ZHANG, A.Q., ZHANG, J.S., TANG, C.H., FAN, J.M., SHI, X.M., JI, Z. and PAN, Y.J. 2006. Structural investigation of a novel fucogalactan that contains 3-*O*-methyl rhamnose isolated from the fruiting bodies of the fungus, *Hericium erinaceus*. Carbohydr. Res. *341*, 645–649.

ZHANG, A.Q., SUN, P.L., ZHANG, J.S., TANG, C.H., FAN, J.M., SHI, X.M. and PAN, Y.J. 2007. Structural investigation of novel fucoglucogalactan isolated from the fruiting bodies of the fungus *Hericium erinaceus*. Food Chem. *104*, 451–456.

ZHAO, M.M., YANG, N., YANG, B., JIANG, Y.M. and ZHANG, G.H. 2007. Structural characterization of water-soluble polysaccharides from *Opuntia monacantha* cladodes in relation to their anti-glycated activities. Food Chem. *105*, 1480–1486.

ZHOU, H.P., LIU, W.L., CHEN, Q.H. and WANG, S.R. 1991. Anti-ageing effect of *Hericium erinaceus* polysaccharide. J. China Pharm. Univ. *22*, 86–88.